

LEVEL - VI

SINGLE ANSWER QUESTIONS

1. Let $f(x) = \lim_{n \rightarrow \infty} \frac{(x^2 + 2x + 3 + \sin \pi x)^n - 1}{(x^2 + 2x + 3 + \sin \pi x)^n + 1}$,

then

A) $f(x)$ is continuous and differentiable for all $x \in \mathbb{R}$

B) $f(x)$ is continuous but not differentiable for all $x \in \mathbb{R}$

C) $f(x)$ is discontinuous at infinite number of points.

D) $f(x)$ is discontinuous at finite number of points.

2. Let

$$f(x) = \lim_{h \rightarrow 0} \frac{(\sin(x+h))^{\ln(x+h)} - (\sin x)^{\ln x}}{h}$$

then $f\left(\frac{\pi}{2}\right)$ is

A) equal to 0

B) equal to 1

C) $\ln \frac{\pi}{2}$

D) non existent

3. Let $g(x) = \begin{cases} \frac{x^2 + x \tan x - x \tan 2x}{ax + \tan x - \tan 3x}; & x \neq 0 \\ 0; & x = 0 \end{cases}$.

If $g'(0)$ exists and is equal to non zero value

b, then $\frac{b}{a}$ is equal to

A) $\frac{7}{13}$

B) $\frac{7}{26}$

C) $\frac{7}{52}$

D) $\frac{5}{52}$

4. If $\lim_{n \rightarrow \infty} \frac{n \cdot 3^n}{n(x-2)^n + n \cdot 3^{n+1} - 3^n} = \frac{1}{3}$ where

$n \in \mathbb{N}$, then the number of interger(s) in the range 'x' is

A) 3

B) 4

C) 5

D) infinite

5. Let

$$f(x) = \frac{e^{\tan x} - e^x + \ln(\sec x + \tan x) - x}{\tan x - x}$$

be a continuous function at $x = 0$. The value of $f(0)$ equals

A) $\frac{1}{2}$

B) $\frac{2}{3}$

C) $\frac{3}{2}$

D) 2

- 6.

$$f(x) = \begin{cases} \frac{\cos^{-1}(1 - \{x\}^2) \sin^{-1}(1 - \{x\})}{\sqrt{2}(\{x\} - \{x\}^3)}, & \text{for } x \neq 0 \\ \frac{\pi}{2}, & \text{for } x = 0 \end{cases}$$

where $\{.\}$ denotes fractional part of x , then, at $x = 0$, $f(x)$, is

A) continuous

B) only left continuous

C) only right continuous

D) neither left continuous nor right continuous

7. The value of

$$\lim_{x \rightarrow \frac{\pi}{2}} \left(2^{\frac{1}{\cos^2 x}} + 3^{\frac{1}{\cos^2 x}} + 4^{\frac{1}{\cos^2 x}} + 5^{\frac{1}{\cos^2 x}} + 6^{\frac{1}{\cos^2 x}} \right)^{2 \cos^2 x}$$

is

A) 1

B) 6

C) 36

D) $\frac{1}{36}$

8. If $f(x)$ is continuous function $\forall x \in \mathbb{R}$ and the range of $f(x)$ is $(2, \sqrt{26})$ and

$$g(x) = \left[\frac{f(x)}{c} \right] \text{ is continuous } \forall x \in \mathbb{R},$$

then the least positive integral value of c is (where $[.]$ denotes the greatest integer function)

A) 2

B) 3

C) 5

D) 6

9. The function $f(x)$ defined by

$$f(x) = \begin{cases} \log_{(4x-3)}(x^2 - 2x + 5), & \frac{3}{4} < x < 1 \text{ and } x > 1 \\ 4, & x = 1 \end{cases}$$

A) is continuous at $x = 1$

B) is discontinuous at $x = 1$ since $f(1^+)$ does not exist though $f(1^-)$ exists.

C) is discontinuous at $x = 1$ since $f(1^-)$ does not exist though $f(1^+)$ exists.

D) is discontinuous at $x = 1$ since neither $f(1^+)$ nor $f(1^-)$ exists.

10. $f(x) = \max \left\{ \frac{x}{n}, |\sin \pi x|, n \in N \right\}$ has maximum points of non-differentiability for $x \in (0, 4)$, then

A) maximum value of n is more than 4.5
 B) least value of n is more than 3.5
 C) maximum value of n is less than 4.5
 D) least value of n is less than 3.5

11. $\lim_{x \rightarrow \infty} \frac{\log_{10}(x+2) + [x] + 2}{x + \sum_{r=0}^{\infty} \left(\frac{1}{2}\right)^r}$ (where $[\cdot]$ is

G.I.F.) =

A) 1 B) 0
 C) 2 D) does not exist

12. If $xy > 0$ and $(x+y)\frac{y}{2} = \frac{-3}{2}$, $x+y + \frac{y}{x} = \frac{1}{2}$

then $\lim_{n \rightarrow \infty} \sum_{r=0}^n \left(\frac{x}{y}\right)^r =$

A) 1 B) 3 C) 0 D) $\frac{3}{2}$

13. Let $f: R \rightarrow R$ be a continuous into function satisfying

$f(x) + f(-x) = 0, \forall x \in R$. If $f(-3) = 2$

and $f(5) = 4$ in $[-5, 5]$, then the equation

$f(x) = 0$ has

A) exactly three real roots
 B) exactly two real roots
 C) atleast five real roots
 D) atleast three real roots

14. Let x_n be defined as $\left(1 + \frac{1}{n}\right)^{n+x_n} = e$, then

$\lim_{n \rightarrow \infty} x_n$ equals

A) 1 B) $\frac{1}{2}$ C) $\frac{1}{e}$ D) 0

15. Let $f(x)$ be continuous for all $x \in R$ except at $x = 0$ and

$f'(x) < 0 \forall x \in (-\infty, 0)$;

$f'(x) > 0 \forall x \in (0, \infty)$. Let

$\lim_{x \rightarrow 0^+} f(x) = 3, \lim_{x \rightarrow 0^-} f(x) = 4, f(0) = 5$

Then the image of $(0, 1)$ about the line

$y = \lim_{x \rightarrow 0} f(\cos^3 x - \cos^2 x) = x \lim_{x \rightarrow 0} f(\sin^2 x - \sin^3 x)$ is

A) $\left(\frac{12}{25}, \frac{-9}{25}\right)$ B) $\left(\frac{11}{25}, \frac{-9}{25}\right)$
 C) $\left(\frac{16}{25}, \frac{-8}{25}\right)$ D) $\left(\frac{24}{25}, \frac{-7}{25}\right)$

MULTI ANSWER QUESTIONS

16. Which of the following statements is/are true?

A) If 'f' is differentiable at $x = c$ then

$\lim_{h \rightarrow 0} \frac{f(c+h) - f(c-h)}{2h}$ exists and equals $f'(c)$

B) Given a function 'f' and a point 'c' in the

domain of f, if $\lim_{h \rightarrow 0} \frac{f(c+h) - f(c-h)}{h}$ exists

then the function is differentiable at $x = c$

C) Let $g(x) = \begin{cases} x^2 \sin \frac{1}{x^2}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ then g' exists

D) Let $g(x) = \begin{cases} x \sin \frac{1}{x^2}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ then g' exists

and is continuous.

17. Which of the following function(s) not defined at $x = 0$ has/have removable discontinuity at $x = 0$?

$$A) f(x) = \frac{1}{1 + 2^{\cot x}}$$

$$B) f(x) = \cos\left(\frac{|\sin x|}{x}\right)$$

$$C) f(x) = x \sin\left(\frac{\pi}{x}\right) \quad D) f(x) = \frac{1}{\ln|x|}$$

18. $f(x) = \min\{1, \cos x, 1 - \sin x\}$, $-\pi \leq x \leq \pi$, then

- A) $f(x)$ is not differentiable at 0
 B) $f(x)$ is differentiable at $\pi/2$
 C) $f(x)$ has local maxima at 0
 D) $f(x)$ local maximum at $x = \pi/2$

19. If $f(x) = \min\{\tan x, \cot x\}$, then

A) $f(x)$ is discontinuous at $x = 0, \frac{\pi}{4}, \frac{5\pi}{4}$

B) $f(x)$ is continuous at $x = 0, \frac{\pi}{2}, \frac{3\pi}{2}$

C) Range of $f(x)$ is $(-\infty, -1] \cup [0, 1]$

D) $f(x)$ is periodic with period π

20. The function $f(x) = \|e^x - 1\| - 1$ is

- A) continuous for all x
 B) differentiable for all x
 C) not continuous at $x = 0, \ln 2$
 D) not differentiable at $x = \ln 2$

21. Given a real valued function f such that

$$f(x) = \begin{cases} \frac{\tan^2\{x\}}{(x^2 - [x]^2)} & \text{for } x > 0 \\ 1 & \text{for } x = 0 \\ \sqrt{\{x\} \cot\{x\}} & \text{for } x < 0 \end{cases}, \text{ where } [x] \text{ is}$$

the integral part and $\{x\}$ is the fractional part of x , then

A) $\lim_{x \rightarrow 0^+} f(x) = 1$

B) $\lim_{x \rightarrow 0^-} f(x) = \cot 1$

C) $\cot^{-1}\left(\lim_{x \rightarrow 0^-} f(x)\right)^2 = 1$

D) $\tan^{-1}\left(\lim_{x \rightarrow 0^+} f(x)\right) = \frac{\pi}{4}$

22. If $f(x) = \operatorname{sgn}(x^2 - ax + 1)$ has maximum number of points of discontinuity, then

A) $a \in (2, \infty)$ B) $a \in (-\infty, 2)$

C) $a \in (-2, 2)$ D) $a \in [-2, 2]$

23. If $f(x) = \begin{cases} x^2(\operatorname{sgn}[x] + \{x\}), & 0 \leq x < 2 \\ \sin x + |x - 3|, & 2 \leq x < 4 \end{cases}$,

where $[]$ and $\{\}$ represent the greatest integer and the fractional part function, respectively.

A) $f(x)$ is differentiable at $x = 1$

B) $f(x)$ is continuous but non-differentiable at $x = 1$

C) $f(x)$ is non-differentiable at $x = 2$

D) $f(x)$ is discontinuous at $x = 2$.

24. Given $f(x) = \sum_{r=1}^n (x^r + x^{-r})^2$; $x \neq \pm 1$ and

$$g(x) = \begin{cases} \lim_{n \rightarrow \infty} ((f(x) - 2n)x^{-2n-2}(1-x^2)) & \text{for } x \neq \pm 1 \\ -1, & \text{for } x = \pm 1 \end{cases}$$

then $g(x)$

A) is discontinuous at $x = -1$

B) is continuous at $x = 2$

C) has a removable discontinuity at $x = 1$

D) has an irremovable discontinuity at $x = 1$

COMPREHENSION QUESTIONS

PASSAGE : 1

$$f(x) = \begin{cases} x + a & x < 0 \\ |x - 1| & x \geq 0 \end{cases}$$

$$g(x) = \begin{cases} x + 1 & x < 0 \\ (x - 1)^2 + b & x \geq 0 \end{cases}$$

Where a and b are non-negative real numbers

25. The value of a , if $(g \circ f)(x)$ is continuous for all real x , is

A) -1 B) 0 C) 1 D) 2

26. The value of b , if $(g \circ f)(x)$ is continuous for all real x , is

A) -1 B) 0 C) 1 D) 2

27. For these values of a and b , $(g \circ f)(x)$ for all $x \in (-1, 1)$ is

A) Even B) odd

C) neither even nor odd

D) symmetrical about x -axis

PASSAGE : 2

Two functions $f(x)$ & $g(x)$ are defined as,

$$f(x) = \begin{cases} [h(x)] - \left\{ \frac{h(x)}{2} \right\}, & \text{for } x \in \text{domain of } h \\ 0, & \text{for } x \notin \text{domain of } h \end{cases}$$

$$g(x) = \begin{cases} \operatorname{sgn} h(x), & \text{for } x \in \text{domain of } h \\ 0, & \text{for } x \notin \text{domain of } h \end{cases}$$

Where,

$$h(x) = \frac{1}{\sqrt{b-a}} \cdot \frac{\sqrt{\frac{b-a}{a}} \sin 2x}{\sqrt{1 + \left(\sqrt{\frac{b-a}{a}} \sin x \right)^2}} \cdot \sqrt{a + b \tan^2 x}$$

for $b > a > 0$ & $[.]$ denotes G.I.F & $\{ \}$ denotes fractional part of x . Now answer the following.

28. For $h(x)$ which one of the following is true.

A) $h(x)$ is continuous at $x = 0$ and $x = \frac{\pi}{2}$ B) $h(x)$ is continuous at $x = 0$ butdiscontinuous at $x = \frac{\pi}{2}$

LIMITS, CONTINUITY & DIFFERENTIABILITY

C) $h(x)$ is discontinuous at $x = 0$ butcontinuous at $x = \frac{\pi}{2}$ D) $h(x)$ is discontinuous at $x = 0$ & $x = \frac{\pi}{2}$

29. For $f(x)$, which of the following is true

A) $f(x)$ is continuous at $x = 0$ and $x = \frac{\pi}{2}$ B) $f(x)$ is continuous at $x = 0$ butdiscontinuous at $x = \frac{\pi}{2}$ C) $f(x)$ is discontinuous at $x = 0$ butcontinuous at $x = \frac{\pi}{2}$ D) $f(x)$ is discontinuous at $x = 0$ & $x = \frac{\pi}{2}$

30. $g(\pi/4) + g(2\pi/3) + g(0) + g(\pi/2) =$

A) 0 B) 1 C) -1 D) 2

PASSAGE : 3

$$\text{Let } f(x) = \begin{cases} x+2, & 0 \leq x < 2 \\ 6-x, & x \geq 2 \end{cases}$$

$$g(x) = \begin{cases} 1 + \tan x, & 0 \leq x < \frac{\pi}{4} \\ 3 - \cot x, & \frac{\pi}{4} \leq x < \pi \end{cases}$$

31. $f(g(x))$ is

A) discontinuous at $x = \frac{\pi}{4}$ B) differentiable at $x = \frac{\pi}{4}$ C) continuous but non-differentiable at $x = \frac{\pi}{4}$ D) differentiable at $x = \frac{\pi}{4}$, but derivative is not continuous.

32. The number of points on non-differentiability of $h(x) = |f(g(x))|$ is
A) 1 B) 2 C) 3 D) 4
33. The range of $h(x) = f(g(x))$ is
A) $(-\infty, \infty)$ B) $(4, \infty)$
C) $(-\infty, 4)$ D) $[-4, 5]$

PASSAGE : 4

Let $f(x) = 2 + |x - 1|$ and

$g(x) = \min(f(t))$ where $x \leq t \leq x^2 + x + 1$
then answer the following:

34. Number of points of discontinuity of $g(x)$ is
A) 1 B) 2 C) 3 D) 0
35. Number of points where the function $g(x)$ is not differentiable is
A) 1 B) 2 C) 3 D) 0
36. Range of $g(x)$ is
A) $[2, \infty)$ B) $[2, 2)$ C) $[0, 2]$ D) $(-\infty, \infty)$

PASSAGES : 5

Let A be a $n \times n$ matrix given by

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix} \text{ such that each}$$

horizontal row is an arithmetic progression and each vertical column is a geometrical progression. It is known that each column in geometric progression have the same

common ratio. Given that $a_{24} = 1$, $a_{42} = \frac{1}{8}$

and $a_{43} = \frac{3}{16}$.

37. Let $S_n = \sum_{j=1}^n a_{4j}$, then $\lim_{n \rightarrow \infty} \frac{S_n}{n^2}$ is equal to
A) $\frac{1}{4}$ B) $\frac{1}{8}$ C) $\frac{1}{16}$ D) $\frac{1}{32}$

38. Let d_i be the common difference of the elements in i^{th} row, then $\sum_{i=1}^n d_i$
A) n B) $\frac{1}{2} - \frac{1}{2^{n+1}}$
C) $1 - \frac{1}{2^n}$ D) $\frac{n+1}{2^n}$

39. The value of $\lim_{n \rightarrow \infty} \sum_{i=1}^n a_{ii}$ is equal to

- A) $\frac{1}{4}$ B) $\frac{1}{2}$ C) 1 D) 2

PASSAGE : 6

If $f(x) = x^2 - 2|x|$

$$g(x) = \begin{cases} \text{minimum}\{f(t) : -2 \leq t \leq x, -2 \leq x < 0\} \\ \text{minimum}\{f(t) : 0 \leq t \leq x, 0 \leq x \leq 3\} \end{cases}$$

then

40. The function $y = |f(x)|$ is differentiable for
A) $x \in R$ B) $x \in R - \{0\}$
C) $x \in R - \{0, 2\}$ D) $(-2, 2)$
41. The function $g(x)$ is differentiable for
A) $x \in [-2, 3]$
B) $x \in [-2, 3] - \{-1, 0, 1\}$
C) $x \in [-2, 3] - \{0, 1\}$
D) $x \in [-2, 3] - \{-1, 0, 2\}$
42. $g(x) = \dots$ ($x \in [-1, 0]$)
A) 0 B) $-\frac{1}{2}$ C) -1 D) $7/3$

PASSAGE : 7

Suppose f, g and h be three real valued function defined on R.

Let $f(x) = 2x + |x|$, $g(x) = \frac{1}{3}(2x - |x|)$ and

$$h(x) = f(g(x))$$

43. The range of the function

$k(x) = 1 + \frac{1}{\pi} (\cos^{-1}(h(x)) + \cot^{-1}(h(x)))$ is equal to

- A) $\left[\frac{1}{4}, \frac{7}{4}\right]$ B) $\left[\frac{5}{4}, \frac{11}{4}\right]$ C) $\left[\frac{1}{4}, \frac{5}{4}\right]$ D) $\left[\frac{7}{4}, \frac{11}{4}\right]$

44. The domain of definition of the function

$l(x) = \sin^{-1}(f(x) - g(x))$ is equal to

- A) $\left[\frac{3}{8}, \infty\right)$ B) $(-\infty, 1]$
C) $[-1, 1]$ D) $\left(-\infty, \frac{3}{8}\right]$

45. The function

$T(x) = f(g(f(x))) + g(f(g(x)))$ is

- A) continuous and differentiable in $(-\infty, \infty)$
B) continuous but not derivable $\forall x \in R$
C) neither continuous nor derivable $\forall x \in R$
D) an odd function

PASSAGES : 8

Consider the function $f: R \rightarrow R$, defined

$$\text{as } f(x) = \begin{cases} x^2 - x + 3, & x \in (-\infty, 3) \cap Q \\ x + a, & x \in (-\infty, 2) - Q \\ 2^x + 1, & x \in (2, 3) - Q \\ 9 \tan\left(\frac{\pi x}{12}\right), & x \in [3, 6) \end{cases}$$

46. If $f(x)$ is continuous at $x = 2$ then the value of a is

- A) 1 B) 2
C) 3 D) indeterminate

47. The function $f(x)$ at $x = 3$

- A) has non-removable discontinuity
B) has removable discontinuity
C) is differentiable
D) is continuous but not differentiable

48. $f'(4)$ is equal to

- A) π B) 3π C) $\frac{3\pi}{2}$ D) $\frac{3\pi}{16}$

MATRIX MATCHING QUESTIONS

49. Column I

LIMITS, CONTINUITY & DIFFERENTIABILITY

A) $f(x) = \begin{cases} \frac{1}{|x|} & \text{for } |x| \geq 1 \\ ax^2 + b & \text{for } |x| < 1 \end{cases}$ is

differentiable every where and $|k| = a + b$, then value of k is

B) If $f(x) = \text{sgn}(x^2 - ax + 1)$ has exactly one point of discontinuity, then the value of a can be

C) $f(x) = [2 + 3|n| \sin x]$, $n \in N$, $x \in (0, \pi)$ has exactly 11 points of discontinuity, then the value of n is

D) $f(x) = ||x| - 2 + a|$ has exactly three points of non-differentiability, then the value of a is

Column II

- P) 2
Q) -2
R) 1
S) -1

50. Match the following
Column I

A) If $f(x) = \frac{1 - \cos(1 - \cos x)}{x^4}$ is

continuous every where $f(0) = \frac{1}{\lambda}$

and let μ be the number of points

$g(x) = \max(x^2 - 1, 4, x^2 + 1)$ is non-

differentiable, then $\lambda + \mu$ is divisible by

B) 'f' be a continuous function on R. If

$f\left(\frac{1}{3^n}\right) = (\cos e^n) 3^{-n^2} + \frac{n^2}{n^2 + n + 1}$ and

$f(0) = \frac{1}{\lambda}$ and let μ be the number of

points if $g(x) = \begin{cases} \min\{x, x^2\}, & x \geq 0 \\ \min\{2x, x^2 - 1, x < 0\} \end{cases}$ is

non differentiable in $[-2, 2]$, then $\lambda + \mu$ is divisible by

C) If $f(x) = \frac{\sqrt{2} \cos x - 1}{\cot x - 1}$ is continuous at

$x = \frac{\pi}{4}$ and $f\left(\frac{\pi}{4}\right) = \frac{1}{\lambda}$ and μ be the jump of the function at the point of discontinuity

of the function $g(x) = \frac{1 - k^{1/x}}{1 + k^{1/x}} (k > 0)$, then

$|\lambda - \mu|$ is divisible by

Column II

P) 2

Q) 3

R) 4

S) 5

51. Let $f(x)$ be a real valued function

defined by $f(x) = x^2 - 2|x|$ and

$$g(x) = \begin{cases} \text{minimum}\{f(t) : -2 \leq t \leq x\}, & x \in [-2, 0) \\ \text{maximum}\{f(t) : 0 \leq t \leq x\}, & x \in [0, 3] \end{cases}$$

Column I

A) $g(x)$ is not continuous at x equal to

B) $g(x)$ is not derivable at x equal to

C) Number of integral critical points of $g(x)$ is equal to

D) Absolute maximum value of $g(x)$ is equal to

Column II

P) -2

Q) 0

R) 1

S) 2

T) 3

52. $L = \lim_{x \rightarrow 0} \left[\frac{n_1 \sin x}{x} \right], M = \lim_{x \rightarrow 0} \left[\frac{n_2 x}{\sin x} \right]$

$$N = \lim_{x \rightarrow 0} \left[\frac{n_3 \tan x}{x} \right] \text{ and } P = \lim_{x \rightarrow 0} \left[\frac{n_4 x}{\tan x} \right]$$

where $[\cdot]$ denotes the greatest integer

function and $n_1, n_2, n_3, n_4 \in N$.

Column - I

A) For $n_1 = 3, n_4 = 9$ then P/L is divisible by

B) For $n_2 = 12, n_3 = 2$, then M/N is divisible by

C) For $n_2 = 3, n_4 = 10$ then P+M is divisible by

D) For $n_1 = 4, n_2 = 5, n_3 = 6, n_4 = 7$ then L+N+P+M is divisible by

Column - II

P) 2

Q) 3

R) 4

S) 5

T) 6

INTEGER QUESTIONS

53. If $\lim_{x \rightarrow 0} \frac{1 - \cos\left(1 - \cos \frac{x}{2}\right)}{2^m x^n}$ is equal to the left hand derivative of $e^{-|x|}$ at $x = 0$, then find the value of $(n^2 + m)$.

54. If $f(x) = \begin{cases} \text{sgn}(x-2) \times [\log_e x], & 1 \leq x \leq 3 \\ \{x^2\}, & 3 < x \leq 3.5 \end{cases}$,

where $[\cdot]$ denotes the greatest integer function and $\{\cdot\}$ represents the fractional part function, then the number of points of discontinuity is.....

55. Let $\lim_{x \rightarrow 1} \left[\frac{x^a - ax + a - 1}{(x-1)^2} \right] = f(a)$. Then the value of $f(4)$ is

56. If f and g are two differentiable functions with $g'(a) = 2$, $g(a) = b$, such that fog = identity function, then $2f'(b)$ is

57. If $f(x), g(x)$ be twice differentiable function on $[0, 2]$ satisfying

$$f''(x) = g''(x), f'(1) = 2, g'(1) = 4 \text{ and}$$

$f(2) = 3, g(2) = 9$, then $-f(x) + g(x)$ at $x = 2$ equals

58. Let S denote sum of the series

$$\frac{3}{2^3} + \frac{4}{2^4 \cdot 3} + \frac{5}{2^6 \cdot 3} + \frac{6}{2^7 \cdot 5} + \dots \infty.$$

Then the value of S^{-1} is

59. The largest value of the non-negative integer a for which

$$\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4} \text{ is}$$

60. The value of

$$\left[\lim_{n \rightarrow \infty} \left(2 \times 3^2 \times 2^3 \times 3^4 \dots 2^{n-1} \times 3^n \right)^{\frac{1}{(n^2+1)}} \right]^4 \text{ is}$$

61. If $L = \lim_{x \rightarrow 0} \frac{\sin x^4 - x^4 \cos x^4 + x^{20}}{x^4 (e^{2x^4} - 1 - 2x^4)}$, then the value of $1/L$ is _____

62. If $\lim_{x \rightarrow 0} \sin \left(\frac{\pi(1 - \cos^m x)}{x^n} \right)$ exists, where $m, n \in \mathbb{N}$, then the sum of all possible values of n is _____.

63. If $\lim_{x \rightarrow \infty} (e^{2x} + e^x + x)^{\frac{1}{x}} = e^a$, then the value of a is

64. If a and b are the numbers of points of non differentiability of $f(x) = [\sin^{-1} x]$

$$\text{and } f(x) = \left[\frac{2}{1+x^2} \right], x \geq 0, (\text{where } [.] \text{ represents greatest integer function})$$

respectively, then the value of $a+b$ is

65. The least integral value of a for which the function

$$f(x) = \left[(x-2)^3 / a \right] \sin(x-2) + a \cos(x-2),$$

where $[.]$ denotes the greatest integer

LIMITS, CONTINUITY & DIFFERENTIABILITY

function, is continuous in $[0,2]$ is _____

66. If a is the number of points of continuity

$$\text{of } f(x) = \begin{cases} x-1, & x \text{ is rational} \\ x^2-x-2, & x \text{ is irrational} \end{cases},$$

b is the number of points of discontinuity

of $f(x) = \operatorname{sgn}(x^3 - 3x + 1)$, c is the

number of points of non-differentiability

of $f(x) = (\log x) |x^2 - 4x + 3| + 2(x-2)^{1/3}$,

and d is the number of points where the graph of

$$f(x) = (\log x) |x^2 - 4x + 3| + 2(x-2)^{1/3} \text{ has}$$

a sharp turn then the value of

$$a+b+c+d \text{ is } \underline{\hspace{2cm}}.$$

67. The number of points of discontinuity of

$$f(x) = \lim_{n \rightarrow \infty} \left[(x^{2n} - 1) / (x^{2n} + 1) \right] \text{ is } \underline{\hspace{2cm}}$$

68. Given the continuous function

$$f(x) = \begin{cases} 2x+3 & x \leq 1 \\ -x^2+6 & x > 1 \end{cases}, \text{ then number of}$$

points where $|f(x)|$ is non-differentiable is _____.

KEY - LEVEL - VI

SINGLE ANSWER QUESTIONS

- | | | | |
|-------|-------|-------|-------|
| 1. A | 2. A | 3. C | 4. C |
| 5. C | 6. C | 7. C | 8. D |
| 9. D | 10. B | 11. A | 12. B |
| 13. D | 14. B | 15. D | |

MULTIANSWER QUESTIONS

- | | |
|-------------|----------------|
| 16. A, C | 17. B, D |
| 18. A, C | 19. C, D |
| 20. A, D | 21. A, B, C, D |
| 22. A, B | 23. A, C, D |
| 24. A, B, D | |

COMPREHENSION QUESTIONS

- | | | | |
|-------|-------|-------|-------|
| 25. C | 26. B | 27. A | 28. B |
| 29. D | 30. A | 31. C | 32. B |
| 33. C | 34. C | 35. C | 36. A |

37. D 38. C 39. D 40. B
 41. C 42. C 43. B 44. D
 45. B 46. C 47. D 48. B

MATRIX MATCHING QUESTIONS

49. A-R, S; B-P, Q; C-P, Q; D-P, R
 50. A-PS; B-PR, C-PQRS
 51. A-Q, B-Q, S, C-T, D-T
 52. A-PR; B-PQT; C-P, Q, R, T; D-P, S

INTEGER QUESTIONS

53. 9 54. 4 55. 6 56. 1
 57. 6 58. 2 59. 2 60. 6
 61. 6 62. 3 63. 2 64. 5
 65. 9 66. 8 67. 2 68. 5

HINTS-LEVEL - VI**SINGLE ANSWER QUESTIONS**

1. $x^2 + 2x + 3 + \sin \pi x = (x+1)^2 + 2 + \sin \pi x > 1$

$\therefore f(x) = 1 \forall x \in \mathbb{R}$

2. If $g(x) = (\sin x)^{\ln x}$

$$f(x) = g'(x) = (\sin x)^{\ln x} \left[\cot x (\ln x) + \frac{\ln(\sin x)}{x} \right]$$

Hence $f\left(\frac{\pi}{2}\right) = g'\left(\frac{\pi}{2}\right) = 1(0+0) = 0$ (Ans.)

3. $g'(0) = b = \lim_{x \rightarrow 0} \frac{x^2 + x \tan x - x \tan 2x}{x(\alpha + \tan x - \tan 3x)} = \lim_{x \rightarrow 0} \frac{x + \tan x - \tan 2x}{\alpha + \tan x - \tan 3x}$

$$= \lim_{x \rightarrow 0} \frac{x \left(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots \right) - \left(2x + \frac{8x^3}{3} + \frac{2}{15}32x^5 + \dots \right)}{\alpha + \left(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots \right) - \left(3x + \frac{27x^3}{3} + \frac{2}{15}243x^5 + \dots \right)}$$

On simplifying $a = 2, b = \frac{7}{26}$.

4. We have $\lim_{n \rightarrow \infty} \frac{n \cdot 3^n}{n(x-2)^n + n \cdot 3^{n+1} - 3^n} = \frac{1}{3}$;

so $\lim_{n \rightarrow \infty} \frac{1}{\left(\frac{x-2}{3}\right)^n + 3 - \frac{1}{n}} = \frac{1}{3}$

Clearly $-1 < \frac{x-2}{3} < 1 \Rightarrow -1 < x < 5$

$\therefore$ Possible integers in the range 'x' are 0, 1, 2, 3, 4 $\Rightarrow$ 5 integers.

5. For continuity of f at $x = 0$, we have

$$k = f(0) = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{e^{\tan x} - e^x}{\tan x - x} + \lim_{x \rightarrow 0} \frac{\ln(\sec x + \tan x) - x}{\left(\frac{\tan x - x}{x^3}\right)x^3}$$

$$= \lim_{x \rightarrow 0} \frac{e^x (e^{\tan x - x} - 1)}{\tan x - x} + 3 \lim_{x \rightarrow 0} \frac{\ln(\sec x - \tan x) - x}{x^3}$$

$$= 1 + 3 \lim_{x \rightarrow 0} \frac{\sec x - 1}{3x^2} \quad (\text{Using LH Rule})$$

$$= 1 + \frac{1}{2} = \frac{3}{2}$$

6. $\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} \frac{\sin^{-1}(1-h)}{\sqrt{2}(1-h)(1+h)} \cdot \lim_{h \rightarrow 0} \frac{\cos^{-1}(1-h^2)}{h} = \frac{\pi}{2}$.

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} \frac{\cos^{-1} h (2-h) \cdot \sin^{-1} h}{\sqrt{2}(1-h)(2-h)} = \frac{\pi}{4\sqrt{2}}$$

7. $\left(\frac{1}{6^{\cos^2 x}}\right)^{2 \cos^2 x} \cdot \lim_{x \rightarrow \frac{\pi}{2}} \left[\left(\frac{1}{3}\right)^{\frac{1}{\cos^2 x}} + \left(\frac{1}{2}\right)^{\frac{1}{\cos^2 x}} + \left(\frac{2}{3}\right)^{\frac{1}{\cos^2 x}} + \left(\frac{5}{6}\right)^{\frac{1}{\cos^2 x}} + 1 \right] = 36$

8. $\therefore 2 < f(x) < \sqrt{26}$

$$\therefore \frac{2}{c} < \frac{f(x)}{c} < \frac{\sqrt{26}}{c}$$

since, $g(x) = \left[\frac{f(x)}{c} \right]$ is continuous $\forall x \in \mathbb{R}$

$$\Rightarrow g(x) = 0 \text{ for } c = 6$$

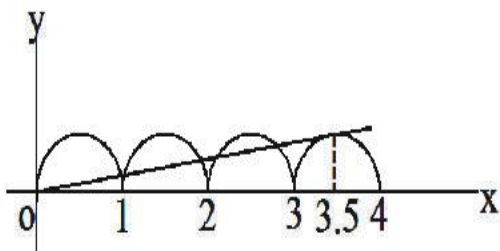
9. We have $\lim_{x \rightarrow 1^-} f(x) = \lim_{h \rightarrow 0} f(1-h)$

$$= \lim_{h \rightarrow 0} \frac{\log(4+h^2)}{\log(1-4h)} = -\infty \quad \text{and}$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} \frac{\log(4+h^2)}{\log(1+4h)} = \infty$$

So, $f(1^-)$ and $f(1^+)$ do not exist.

10. $f(x) = \max \left\{ \frac{x}{n}, |\sin \pi x| \right\}$



Thus, for the maximum points of non differentiability, graphs of $y = \frac{x}{7}$ and $y = |\sin \pi x|$ must intersect at maximum number of points which occurs when $n > 3.5$. Hence, the least value of n is 4.

11. $\lim_{x \rightarrow \infty} \frac{\log_{10}(x+2) + [x] + 2}{[x] + 2}$

$$= 1 + \lim_{x \rightarrow \infty} \frac{\log_{10}(x+2)}{[x] + 2}$$

$$= 1 + \lim_{x \rightarrow \infty} \frac{\log_{10}(x+2)}{\frac{x}{x} + \frac{2}{x}}$$

$$= 1 + 0$$

$$= 1$$

12. Let $\frac{y}{x} = a \Rightarrow x + y = \frac{-3}{2a}, x + y = \frac{1}{2} - a$

$$-\frac{3}{2a} = \frac{1}{2} - a \Rightarrow -\frac{3}{2a} = \frac{1-2a}{2}$$

$$\Rightarrow -3 = a - 2a^2 \Rightarrow 2a^2 - a - 3 = 0$$

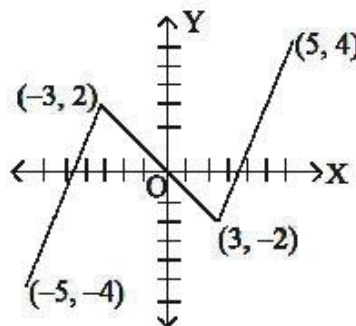
$$\Rightarrow a = -1 \text{ or } a = \frac{3}{2} \quad \therefore \frac{x}{y} = \frac{2}{3}$$

$$\therefore \lim_{n \rightarrow \infty} \sum_{r=0}^n \left(\frac{x}{y} \right)^r = 3$$

13. $f(x) + f(-x) = 0 \Rightarrow f(x)$ is an odd function. Since points $(-3, 2)$ and $(5, 4)$ lie on the curve

$\therefore (3, -2)$ and $(-5, -4)$ will also lie on the curve.

For minimum number of roots, graph of continuous function $f(x)$ is as follows.



from the above two graphs of $f(x)$ it is clear that equation $f(x) = 0$ has atleast three real roots.

14. Given $\left(1 + \frac{1}{n}\right)^{n+x_n} = e$

taking log

$$(n+x_n) \ln \left(1 + \frac{1}{n}\right) = 1 \Rightarrow n+x_n = \frac{1}{\ln \left(1 + \frac{1}{n}\right)}$$

$$\Rightarrow x_n = \frac{1}{\ln \left(1 + \frac{1}{n}\right)} - n \quad \dots (1)$$

$$\text{let } \frac{n+1}{n} = u \Rightarrow nu = n+1 \Rightarrow n = \frac{1}{u-1}$$

$$x_n = \lim_{u \rightarrow 1} \left(\frac{1}{\ln u} - \frac{1}{u-1} \right) = \lim_{u \rightarrow 1} \frac{(u-1) - \ln u}{(u-1) \ln u}$$

$$= \lim_{u \rightarrow 1} \frac{1 - \frac{1}{u}}{\frac{u-1}{u} + \ln u} = \lim_{u \rightarrow 1} \frac{\frac{u^2 - 1}{u^2}}{\frac{u-1}{u^2} + \frac{1}{u}} = \frac{1}{2}$$

15. $\cos^3 x - \cos^2 x \rightarrow 0$ from LHS
 $\sin^2 x - \sin^3 x \rightarrow 0$ from RHS
 $\therefore$ the given line reduces to $4y = 3x$

MULTIANSWER TYPE QUESTIONS

16. a) $\lim_{h \rightarrow 0} \frac{f(c+h) - f(c) + f(c) - f(c-h)}{2h}$
 $= \frac{1}{2} f'(c) + \frac{1}{2} f'(c) = f'(c)$
 b) existence of limit cannot confirm differentiability
17. (b,c,d)

(a) $f(0^-) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{1}{1 + 2^{\cot(0-h)}}$
 $= \lim_{h \rightarrow 0} \frac{1}{1 + 2^{-\cot h}} = \frac{1}{1 + 2^{-\infty}} = \frac{1}{1+0} = 1$
 $\therefore f(0^-) \neq f(0^+)$

(b)

$$f(0^-) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \cos \left(\frac{|\sin(0-h)|}{0-h} \right)$$

$$= \lim_{h \rightarrow 0} \cos \left(\frac{\sinh}{-h} \right)$$

$$= \cos \left(-\lim_{h \rightarrow 0} \cos \frac{\sinh}{-h} \right) = \cos(-1) = \cos 1$$

and

$$f(0^+) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \cos \left(\frac{|\sin(0-h)|}{0-h} \right)$$

$$= \lim_{h \rightarrow 0} \cos \left(\frac{\sinh}{h} \right) = \cos \left(\lim_{h \rightarrow 0} \frac{\sinh}{h} \right) = \cos 1$$

$$\therefore f(0^-) = f(0^+) \neq f(0)$$

(C) : $f(0^-) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} (-h) \sin \left(\frac{\pi}{-h} \right)$
 $= \lim_{h \rightarrow 0} h \sin \left(\frac{\pi}{h} \right)$

$$= 0 \times \sin \infty$$

$$= 0 \times (\text{lies between } -1 \text{ to } 1)$$

and $f(0^+) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} h \sin \left(\frac{\pi}{h} \right)$
 $= 0 \times \sin \infty = 0 \times (\text{lies between } -1 \text{ to } 1) = 0$
 $\therefore f(0^-) = f(0^+) \neq f(0)$

(d) : $f(0^-) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{1}{\ln|0-h|}$

$$= \lim_{h \rightarrow 0} \frac{1}{\ln h}$$

$$= \frac{1}{\ln 0} = \frac{1}{-\infty} = 0$$

and $f(0^+) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \frac{1}{\ln|0+h|} = \lim_{h \rightarrow 0} \frac{1}{\ln h}$
 $= \frac{1}{-\infty} = 0$

$$\therefore f(0^-) = f(0^+) \neq f(0)$$

18. We have, $f(x) = \min \{1, \cos x, 1 - \sin x\}$
 $\therefore f(x)$ can be rewritten as

$$f(x) = \begin{cases} \cos x, & -\frac{\pi}{2} \leq x \leq 0 \\ 1 - \sin x, & 0 < x \leq \frac{\pi}{2} \\ \cos x, & \frac{\pi}{2} < x \leq \pi \end{cases}$$

$$\Rightarrow f'(x) = \begin{cases} -\sin x, & -\frac{\pi}{2} \leq x \leq 0 \\ -\cos x, & 0 < x \leq \frac{\pi}{2} \\ -\sin x, & \frac{\pi}{2} < x \leq \pi \end{cases}$$

$$\therefore f'(0) = 0$$

hence, $f(x)$ has local maxima at 0 and $f(x)$ is not differentiable at $x = 0$.

19. $f(x)$ is discontinuous at $\frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \dots$

and $f(x)$ is periodic with period π

Range of $f(x)$ is $(-\infty, -1] \cup [0, 1]$

$$20. \quad f(x) = \begin{cases} e^x & x < 0 \\ 2 - e^x & 0 \leq x < \ln 2 \\ e^x - 2 & x \geq \ln 2 \end{cases}$$

f is continuous $\forall x \in R$, but is not differentiable at $x = 0, \ln 2$

$$21. \quad \text{We have } \lim_{k \rightarrow 0^+} f(x) = \lim_{k \rightarrow 0^+} \frac{\tan^2 \{x\}}{(x^2 - [x]^2)}$$

$$= \lim_{x \rightarrow 0^+} \frac{\tan^2 x}{x^2} = 1 \quad \dots(1)$$

$$[\because x \rightarrow 0^+; [x] = 0 = \{x\} = x]$$

$$\text{Also } \lim_{k \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \sqrt{\{x\} \cot \{x\}} = \sqrt{\cot 1} \quad \dots(2)$$

$$[\because x \rightarrow 0^-; [x] = -1 \Rightarrow \{x\} = x + 1 \Rightarrow \{x\} \rightarrow 1]$$

$$\text{Also, } \cot^{-1} \left(\lim_{x \rightarrow 0^-} f(x) \right)^2 = \cot^{-1} (\cot 1) = 1$$

22. For maximum points of discontinuity of $f(x) = \text{sgn}(x^2 - ax + 1)$, $x^2 - ax + 1 = 0$ must have two distinct roots, for which

$$D = a^2 - 4 > 0 \Rightarrow a \in (-\infty, -2) \cup (2, \infty).$$

23. For continuity at $x = 1$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x^2 \text{sgn}[x] + \{x\}) = 1 + 0 = 1$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x^2 \text{sgn}[x] + \{x\}) = 1 \text{sgn}(0) + 1 = 1$$

$$\text{Also, } f(1) = 1$$

$$\therefore \text{L.H.L} = \text{R.H.L} = f(1).$$

Hence $f(x)$ is continuous at $x = 1$.

Now for differentiability,

$$f'(1^+) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

LIMITS, CONTINUITY & DIFFERENTIABILITY

$$= \lim_{h \rightarrow 0} \frac{(1+h)^2 \text{sgn}[1+h] + \{1+h\} - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(1+h)^2 + h - 1}{h} = \lim_{h \rightarrow 0} \frac{h^2 + 3h}{h} = 3$$

$$\text{and } f'(1^-) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{(1-h)^2 \text{sgn}[1-h] + \{1-h\} - 1}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{(1-h)^2 + 1 - h - 1}{-h} = \lim_{h \rightarrow 0} \frac{h^2 - 3h}{-h} = 3$$

$$f'(1^+) = f'(1^-)$$

Hence, $f(x)$ is differentiable at $x = 1$.

Now at $x = 2$,

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (x^2 \text{sgn}[x] + \{x\}) = 4 \times 0 + 1$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (\sin x + |x - 3|) = 1 + \sin 2$$

Hence $LHL \neq RHL$

Hence, $f(x)$ is discontinuous at $x = 2$ and

then $f(x)$ is also non differentiable at $x = 2$.

$$24. \quad f(x) = \sum_{r=1}^n \left(x^{2r} + \frac{1}{x^{2r}} + 2 \right) = \sum_{r=1}^n x^{2r} + \sum_{r=1}^n \frac{1}{x^{2r}} + 2n$$

$$= (x^2 + x^4 + \dots + x^{2n}) + \left(\frac{1}{x^2} + \frac{1}{x^4} + \dots + \frac{1}{x^{2n}} \right) + 2n$$

$$= \frac{x^2(1-x^{2n})}{(1-x^2)} + \frac{1}{x^2} \frac{\left(1 - \frac{1}{x^{2n}}\right)}{1 - \frac{1}{x^2}} + 2n$$

$$f(x) = \frac{x^2(1-x^{2n})}{1-x^2} + \frac{(1-x^{2n})}{(1-x^2)x^{2n}} + 2n$$

$$f(x) = \frac{(1-x^{2n})}{1-x^2} \left(x^2 + \frac{1}{x^{2n}} \right) + 2n \quad x \neq \pm 1$$

$$\therefore (f(x) - 2n)(1-x^2) = (1-x^{2n}) \left(x^2 + \frac{1}{x^{2n}} \right)$$

now consider

$$g(x) = \lim_{n \rightarrow \infty} \left((f(x) - 2n)x^{-2n-2} (1-x^2) \right) \text{ for}$$

$x \neq \pm 1$

$$= \lim_{n \rightarrow \infty} (1-x^{2n}) \left(x^2 + \frac{1}{x^{2n}} \right) \cdot \frac{1}{x^{2n+2}}; \quad x \neq \pm 1$$

$$= \lim_{n \rightarrow \infty} \frac{(1-x^{2n})(x^{2n+2} + 1)}{(x^{2n})(x^{2n+2})}$$

$$= \lim_{n \rightarrow \infty} \left(-1 + \frac{1}{x^{2n}} \right) \left(1 + \frac{1}{x^{2n+2}} \right)$$

$$\text{now } g(x) = \begin{cases} -1 & \text{if } |x| > 1 \\ -\infty & \text{if } |x| < 1 \\ 0 & \text{if } x = \pm 1 \end{cases}$$

now clearly g has non removable infinite type of discontinuity at $x = 1$ and -1

g is continuous at $x = 2$

COMPREHENSION QUESTIONS

PASSAGE : 1 25-27

$\therefore (gof)(x)$ is continuous $g(x)$ and $f(x)$

are also continuous, $\lim_{x \rightarrow 0^-} f(x) = f(0)$

$$\lim_{h \rightarrow 0} (0-h) + a = 1 \Rightarrow a = 1$$

$$\lim_{x \rightarrow 0^-} g(x) = g(0), \quad \lim_{h \rightarrow 0} 0-h+1 = (0-1)^2 + b$$

$$\Rightarrow 1 = 1 + b \Rightarrow b = 0$$

$$(gof)(x) = g(f(x)) = \begin{cases} f(x)+1, & f(x) < 0 \\ (f(x)-1)^2, & f(x) \geq 0 \end{cases}$$

$$= \begin{cases} x+2, & x < -1 \\ x^2, & -1 \leq x < 1 \\ (x-2)^2, & x \geq 1 \end{cases}$$

$\therefore (gof)(x)$ is even for all $x \in (-1, 1)$

PASSAGE : 2 Q.No. 28 to 30

Conceptual

PASSAGE : 3 Q.No. 31 to 33

$$\text{For } 0 \leq x < \frac{\pi}{4}, \quad g(x) = 1 + \tan x$$

$$x \in \left[0, \frac{\pi}{4} \right) \Rightarrow 1 + \tan x \in [1, 2)$$

$$\text{So } f(g(x)) = f(1 + \tan x) = 1 + \tan x + 2$$

$$\text{and for } x \in \left[\frac{\pi}{4}, \pi \right), \quad g(x) = 3 - \cot x$$

$$x \in \left[\frac{\pi}{4}, \pi \right) \Rightarrow 3 - \cot x \in [2, \infty)$$

$$\text{So } f(g(x)) = f(3 - \cot x) = 6 - (3 - \cot x)$$

Let

$$h(x) = f(g(x)) = \begin{cases} 3 + \tan x, & 0 \leq x < \frac{\pi}{4} \\ 3 + \cot x, & \frac{\pi}{4} \leq x < \pi \end{cases}$$

Clearly, $f(g(x))$ is continuous in $[0, \pi)$

$$\text{Now } h'\left(\frac{\pi^+}{4}\right) = \lim_{x \rightarrow \frac{\pi}{4}^+} (-\operatorname{cosec}^2 x) = -2$$

$$h'\left(\frac{\pi^-}{4}\right) = \lim_{x \rightarrow \frac{\pi}{4}^-} (\sec^2 x) = 2$$

So $f(g(x))$ is differentiable everywhere in

$[0, \pi)$ other than at $x = \frac{\pi}{4}$.

$$|f(g(x))| = \begin{cases} |3 + \tan x|, & 0 \leq x < \frac{\pi}{4} \\ |3 + \cot x|, & \frac{\pi}{4} \leq x < \pi \end{cases}$$

Which is non differentiable at $x = \frac{\pi}{4}$ and

where $3 + \cot x = 0$ or $x = \cot^{-1}(-3)$

For $x \in \left[0, \frac{\pi}{4}\right), 3 + \tan x \in [3, 4)$

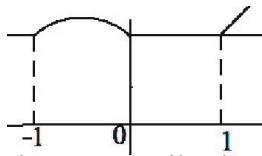
For $x \in \left[\frac{\pi}{4}, \pi\right), 3 + \cot x \in (-\infty, 4]$

Hence, the range is $(-\infty, 4]$.

PASSAGE : 4 Q.No. 34 to 36

$$f(t) = \begin{cases} 3-t, & t \leq 1 \\ t+1, & t > 1 \end{cases}$$

$$g(x) = \begin{cases} 3-(x^2+x+1) & x \in [-1, 0) \\ 2 & x \in (-\infty, -1) \cup (0, 1] \\ x+1 & x \in [1, \infty) \end{cases}$$



From graph we can easily give the answer

PASSAGES : 5 (Q.No. 37 to 39)

$$a_{43} = a_{42} + d_4 \Rightarrow d_4 = \frac{1}{16}$$

(common difference of 4th row)

$$a_{41} = a_{42} - d_4 = \frac{1}{16}$$

$$\therefore a_{41} = \frac{1}{16}, a_{42} = \frac{2}{16}, a_{43} = \frac{3}{16}, a_{44} = \frac{4}{16},$$

$$\dots\dots, a_{4n} = \frac{n}{16}$$

now all elements of 4th row are known

$$S_n = \sum_{j=1}^n a_{4j} = \frac{n(n+1)}{2(16)};$$

$$\lim_{n \rightarrow \infty} \frac{S_n}{n^2} = \frac{1}{32} \text{ (Ans: D)}$$

$$a_{24}r^2 = a_{44} = \frac{4}{16} \Rightarrow r^2 = \frac{1}{4} \Rightarrow r = \frac{1}{2}$$

(common ratio for all G.P. is $\frac{1}{2}$)

note that all the elements of 4th row are known

and common ratio $\frac{1}{2}$ is also known therefore

LIMITS, CONTINUITY & DIFFERENTIABILITY

$(m \times n)$ matrix is

$$A = \begin{bmatrix} \frac{1}{2} & \frac{2}{2} & \frac{3}{2} & \frac{4}{2} & \dots & \frac{n}{2} \\ \frac{1}{2^2} & \frac{2}{2^2} & \frac{3}{2^2} & \frac{4}{2^2} & \dots & \frac{n}{2^2} \\ \frac{1}{2^3} & \frac{2}{2^3} & \frac{3}{2^3} & \frac{4}{2^3} & \dots & \frac{n}{2^3} \\ \frac{1}{2^4} & \frac{2}{2^4} & \frac{3}{2^4} & \frac{4}{2^4} & \dots & \frac{n}{2^4} \\ \frac{1}{2^n} & \frac{2}{2^n} & \frac{3}{2^n} & \frac{4}{2^n} & \dots & \frac{n}{2^n} \end{bmatrix}$$

constructed.

$$\text{Working: } a_{41} = \frac{1}{2^4}, a_{42} = \frac{2}{2^4} \text{ ||ly } a_{4i} = \frac{i}{2^4}$$

$$\text{now } \left. \begin{matrix} a_{11} \\ a_{21} \\ a_{31} \\ a_{41} \end{matrix} \right\} \text{ in G.P. } \Rightarrow \frac{a_{41}}{a_{11}} = \frac{1}{16} = \left(\frac{1}{2}\right)^4 \Rightarrow a_{11} = \frac{1}{2}$$

$$\text{again } a_{42} = \frac{2}{2^4}$$

$$\left. \begin{matrix} a_{12} \\ a_{22} \\ a_{32} \\ a_{42} \end{matrix} \right\} \text{ in G.P. } \Rightarrow \frac{a_{42}}{a_{12}} = \frac{2}{16} = \left(\frac{1}{2}\right)^4 \Rightarrow a_{12} = \frac{1}{2^2}$$

$$\sum_{i=1}^n d_i = \sum_{i=1}^n \frac{1}{2^i} = \frac{1 \left(1 - \frac{1}{2^n}\right)}{\left(1 - \frac{1}{2}\right)} = 1 - \frac{1}{2^n}$$

$$\sum_{i=1}^n a_{ii} = S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \frac{4}{2^4} + \dots + \frac{n}{2^n}$$

$$\frac{1}{2}S = \frac{1}{2^2} + \frac{2}{2^3} + \frac{3}{2^4} + \dots + \frac{n-1}{2^n} + \frac{n}{2^{n+1}}$$

$$\frac{1}{2}S = \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^n} \right) - \frac{n}{2^{n+1}}$$

$$S = 2 \left(1 - \frac{1}{2^n} \right) - \frac{n}{2^n}$$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n a_{ii} = \lim_{n \rightarrow \infty} \left(2 - \frac{2}{2^n} - \frac{n}{2^n} \right)$$

$$= 2 - 0 - 0 \Rightarrow \lim_{n \rightarrow \infty} \sum_{i=1}^n a_{ii} = 2$$

PASSAGE : 6 (q.no. 40 to 42)

$$f(x) \text{ if } -2 \leq x \leq -1$$

$$g(x) = \begin{cases} -1 & \text{if } -1 \leq x < 0 \\ 0 & \text{if } -\leq x < 1 \end{cases}$$

$$f(x) \text{ if } 1 \leq x \leq 3$$

PASSAGE : 7 (q.no. 43 to 45)

We have $f(x) = \begin{cases} 3x, & x \geq 0 \\ x, & x < 0 \end{cases}$ and

$$g(x) = \begin{cases} \frac{x}{3}, & x \geq 0 \\ x, & x < 0 \end{cases}$$

Clearly f and g are inverse of each other

Now, $h(x) = f(g(x)) = \begin{cases} 3\left(\frac{x}{3}\right) = x, & x \geq 0 \\ x, & x < 0 \end{cases}$

(i) As $h(x) = x \forall x \in R$

$$\Rightarrow k(x) = 1 + \frac{1}{\pi} (\cos^{-1} x + \cot^{-1} x)$$

Domain of $k(x) = [-1, 1]$ and $k(x)$ is decreasing function on $[-1, 1]$

As $k(x)$ is continuous function on $[-1, 1]$

Now, $k_{\min}(x=1) = 1 + \frac{1}{\pi} (\cos^{-1} 1 + \cot^{-1} 1)$

$$= 1 + \frac{1}{\pi} \left(0 + \frac{\pi}{4}\right) = 1 + \frac{1}{4} = \frac{5}{4}$$

$$k_{\max}(x=-1) = 1 + \frac{1}{\pi} (\cos^{-1}(-1) + \cot^{-1}(-1))$$

$$= 1 + \frac{1}{\pi} \left(\pi + \frac{3\pi}{4}\right) = 1 + \frac{7}{4} = \frac{11}{4}$$

$$\Rightarrow \text{Range of } k(x) = \left[\frac{5}{4}, \frac{11}{4}\right]$$

(ii) We have

$$f(x) - g(x) = (2x + |x|) - \frac{1}{3}(2x - |x|)$$

$$= \frac{4x}{3} + \frac{4}{3}|x| = \begin{cases} \frac{8}{3}x; & x \geq 0 \\ 0; & x < 0 \end{cases}$$

$\therefore$ For domain of function,

$$0 \leq \frac{8x}{3} \leq 1 \Rightarrow 0 \leq x \leq \frac{3}{8}$$

$$\Rightarrow \text{Domain of } l(x) = \left(-\infty, \frac{3}{8}\right]$$

Note : Range of function $l(x) = \left[0, \frac{\pi}{2}\right]$

(iii) As f and g are inverse of each other, so $T(x) =$

$$T0x = f(g(f(x))) + g(f(g(x)))$$

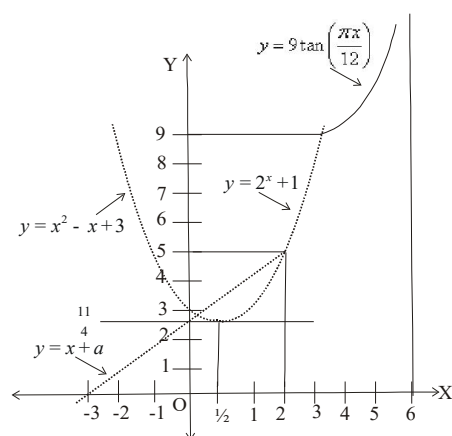
$$= f(x) + g(x) = (2x + |x|) + \frac{1}{3}(2x - |x|)$$

$$\Rightarrow T(x) = \begin{cases} \frac{10x}{3}, & x \geq 0 \\ 2x, & x < 0 \end{cases} \quad \text{Clearly, } T(x) \text{ is}$$

continuous but non-derivable at $x = 0$

PASSAGES : 8 (Q.NO. 46 to 48)

$$f(x) = \begin{cases} x^2 - x + 3, & \text{if } x \in Q \text{ in } (-\infty, 3) \\ x + a, & \text{if } x \notin Q \text{ in } (-\infty, 2) \\ 2^x + 1, & \text{if } x \notin Q \text{ in } (2, 3) \\ 9 \tan\left(\frac{\pi x}{12}\right), & \text{if } x \geq 3 \end{cases}$$



through irrational $f(2^-) = 2 + a$

through rational $f(2) = 4 - 2 + 3 = 5$

Hence for continuity at $x = 2$

$$2 + a = 5 \Rightarrow a = 3$$

$$\text{for } x \geq 3, f(x) = 9 \tan \frac{\pi x}{12}$$

$$\Rightarrow f(3^+) = 9; f(3^-) = 2^3 + 1 = 9$$

$\Rightarrow f$ is continuous at $x = 3$ obvious not derivable

$$f'(x) = 9 \sec^2 \left(\frac{\pi x}{12} \right) \frac{\pi}{12}$$

$$f'(4) = 9 \sec^2 \left(\frac{\pi}{3} \right) \frac{\pi}{12} = 3\pi$$

MATRIX MATCHING QUESTIONS

49. A) The given function is clearly continuous at all points except possibly at $x = \pm 1$.

As $f(x)$ is an even function, so we need to check its continuity only at $x = 1$.

$$\Rightarrow \lim_{x \rightarrow 1^-} (ax^2 + b) = \lim_{x \rightarrow 1^+} \frac{1}{|x|} \Rightarrow a = 1 = b = 1$$

Clearly, $f(x)$ is differentiable for all x ,

except possibly at $x = \pm 1$. As $f(x)$ is an even function, so we need to check its differentiability at $x = 1$ only.

$$\lim_{x \rightarrow 1^-} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1^+} \frac{f(x) - f(1)}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{ax^2 + b - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{ax^2 + b - 1}{x - 1} = \lim_{x \rightarrow 1^+} \frac{\frac{1}{x} - 1}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{ax^2 - a}{x - 1} = \lim_{x \rightarrow 1} \frac{-1}{x} \Rightarrow 2a = -1 \Rightarrow a = -\frac{1}{2}$$

Putting $a = -\frac{1}{2}$ in (1)

$$\text{we get } b = \frac{3}{2} \Rightarrow |k| = 1 \Rightarrow k = \pm 1$$

- B) $f(x) = \operatorname{sgn}(x^2 - ax + 1)$ is discontinuous then $x^2 - ax + 1 = 0$ must have only one real root. Hence $a = \pm 2$.
- C) $f(x) = [2 + 3|n| \sin x], n \in N$ has exactly 10 points of discontinuity in $x \in (0, \pi)$.

LIMITS, CONTINUITY & DIFFERENTIABILITY

The required number of points are

$$1 + 2(3|n| - 1) = 6|n| - 1 = 11 \Rightarrow n = \pm 2$$

- D) If $f(x) = ||x| - 2| + a$ has exactly three points of non differentiability.

$f(x)$ is non-differentiable at $x = 0, |x| - 2 = 0$ or $x = 0, \pm 2$.

Hence, the value of a must be positive, as

negative value of a allows $||x| - 2| + a = 0$ to have real roots, which gives to more points of non differentiability.

50. (A) $\rightarrow (p, s); (B) \rightarrow (p, r); (C) \rightarrow (p, q, r, s)$

$$(A) f(0) = RHL = \lim_{x \rightarrow 0^+} f(x)$$

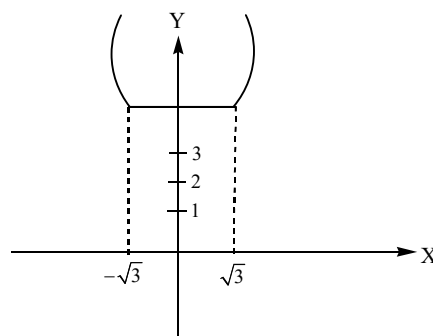
$$= \lim_{h \rightarrow 0} f(0 + h) = \lim_{h \rightarrow 0} f(h)$$

$$= \lim_{h \rightarrow 0} \frac{1 - \cos(1 - \cosh)}{h^4}$$

$$= \lim_{h \rightarrow 0} \frac{1 - \cos(1 - \cosh)}{(1 - \cosh)^2} \cdot \lim_{h \rightarrow 0} \left(\frac{1 - \cosh}{h^2} \right)^2$$

$$= \frac{1}{2} \cdot \left(\frac{1}{2} \right)^2 = \frac{1}{8} = \frac{1}{\lambda} \text{ (given)} \quad \therefore \lambda = 8$$

$$\text{and } g(x) = \max \{x^2 - 1, 4, x^2 + 1\}$$



It is clear from graph number of points of non-differentiable is 2. $\therefore \mu = 2$

then $\lambda + \mu = 10(p, s)$ is divisible by 2 and 5.

$$(B) f(0) = \lim_{n \rightarrow \infty} \left((\cos e^n) 3^{-n^2} + \frac{n^2}{n^2 + n + 1} \right)$$

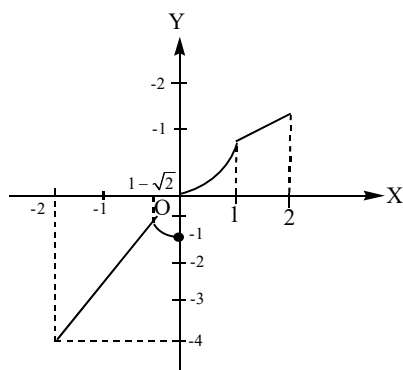
$$= \lim_{n \rightarrow \infty} \left(\frac{(\cos e^n)}{3^{n^2}} + \frac{1}{1 + \frac{1}{n} + \frac{1}{n^2}} \right)$$

$$= \frac{\cos e^{\infty}}{3^{\infty}} + \frac{1}{1+0+0} = \frac{-1 \text{ to } 1}{\infty} + 1 = 0 + 1 = 1$$

$$\therefore \lambda = \frac{1}{f(0)} = 1$$

$$\text{and } g(x) = \begin{cases} \min.\{x, x^2\}, & x \geq 0 \\ \min.\{2x, x^2 - 1\}, & x < 0 \end{cases}$$

$$= \begin{cases} 2x & , -2 \leq x < 1 - \sqrt{2} \\ x^2 - 1 & , 1 - \sqrt{2} \leq x < 0 \\ x^2 & , 0 \leq x < 1 \\ x & , 1 \leq x \leq 2 \end{cases}$$



It is clear from the graph $g(x)$ is non-differentiable at 3 points.

$\therefore \mu = 3$, then $\lambda + \mu = 4(p, r)$ is divisible by 2 and 4.

(C) $\therefore$

$$f\left(\frac{\pi}{4}\right) = RHL = \lim_{x \rightarrow \frac{\pi}{4}^+} f(x) = \lim_{h \rightarrow 0} f\left(\frac{\pi}{4} + h\right)$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{2} \cos\left(\frac{\pi}{4} + h\right) - 1}{\cot\left(\frac{\pi}{4} + h\right) - 1}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{2} \left(\frac{1}{\sqrt{2}} \cosh - \frac{1}{\sqrt{2}} \sinh \right)}{\left(\frac{1 - \tanh}{1 + \tanh} \right) - 1}$$

$$= \lim_{h \rightarrow 0} \frac{\left(\frac{-(1 - \cosh)}{h} - \frac{\sinh}{h} \right) (1 + \tanh)}{-2 \left(\frac{\tanh h}{h} \right)}$$

$$= \frac{(-0-1)(1+0)}{-2(1)} = \frac{1}{2} = \frac{1}{\lambda} \text{ (given)}$$

$$\therefore \lambda = 2 \text{ and } g(x) = \frac{1 - k^{1/x}}{1 + k^{1/x}} \text{ for } 0 < k < 1$$

check at $x=0$

$$\text{LHL} = \lim_{x \rightarrow 0^-} g(x) = \lim_{h \rightarrow 0} g(0-h)$$

$$= \lim_{h \rightarrow 0} \frac{1 - k^{-1/h}}{1 + k^{-1/h}} = \frac{1-0}{1+0} = 1$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} g(x) = \lim_{h \rightarrow 0} g(0+h) = \lim_{h \rightarrow 0} \frac{1 - k^{1/h}}{1 + k^{1/h}}$$

$$\therefore \text{Jump} = \text{LHL} - \text{RHL} = 1 - (-1) = 2$$

$$\therefore \mu = 2 \Rightarrow |\lambda - \mu| = 0 \text{ (p, q, r, s) divisible by 2, 3, 4 and 5}$$

51. $g(x)$ can be defined as

$$g(x) = \begin{cases} f(x), & -2 \leq x < -1 \\ -1, & -1 \leq x < 0 \\ f(x), & 2 \leq x \leq 3 \end{cases}$$

$$\text{or } g(x) = \begin{cases} x^2 + 2x, & -2 \leq x < -1 \\ -1, & -1 \leq x < 0 \\ x^2 - 2x, & 2 \leq x \leq 3 \end{cases}$$

Clearly $g(x)$ is discontinuous at $x=0$

$g(x)$ is not differentiable at $x=0, 2$

Ab solute maximum value of $g(x) = f(3) = 3$.

52. (A) $\rightarrow (p, r)$; (B) $\rightarrow (p, q, t)$;

(C) $\rightarrow (p, q, r, t)$; (D) $\rightarrow (p, s)$

Since $\frac{\sin x}{x} \rightarrow 1$ as $x \rightarrow 0$ but less than 1.

$\therefore \frac{n_1 \sin x}{x} \rightarrow n_1$ as $x \rightarrow 0$ but less than n_1 .

Similarly,

$\frac{x}{\sin x} \rightarrow 1$ as $x \rightarrow 0$ but more than or equal to 1.

$\therefore \frac{n_2 x}{\sin x} \rightarrow n_2$ as $x \rightarrow 0$ but more than or equal to n_2 .

$$\therefore L = \lim_{x \rightarrow 0} \left[\frac{n_1 \sin x}{x} \right] = n_1 - 1$$

$$\text{and } M = \lim_{x \rightarrow 0} \left[\frac{n_2 x}{\sin x} \right] = n_2$$

Also, $\frac{\tan x}{x} \rightarrow 1$ as $x \rightarrow 0$ but more than 1.

$\therefore \frac{n_3 \tan x}{x} \rightarrow n_3$ as $x \rightarrow 0$ but more than or equal to x_3 .

Similarly,

$\frac{x}{\tan x} \rightarrow 1$ as $x \rightarrow 0$ but less than 1.

$\therefore \frac{n_4 x}{\tan x} \rightarrow n_4$ as $x \rightarrow 0$ but less than n_4 .

$$\Rightarrow N = \lim_{x \rightarrow 0} \left[\frac{n_3 \tan x}{x} \right] = n_3 \text{ and}$$

$$P = \lim_{x \rightarrow 0} \left[\frac{n_4 x}{\tan x} \right] = n_4 - 1$$

(A) For $n_1 = 3, n_4 = 9$

$$\therefore \frac{P}{L} = \frac{n_4 - 1}{n_1 - 1} = \frac{9 - 1}{3 - 1} = 4 \quad (p, r)$$

(B) For $n_2 = 12, n_3 = 2$

$$\therefore \frac{M}{N} = \frac{n_2}{n_3} = \frac{12}{2} = 6 \quad (p, q, t)$$

(C) For $n_2 = 3, n_4 = 10$

$$\therefore P + M = n_4 - 1 + n_2 = 10 - 1 + 3 = 12$$

(D) For $n_1 = 4, n_2 = 5, n_3 = 6, n_4 = 7$

$$\therefore L + N + P - M = n_1 - 1 + n_3 + n_4 - 1 - n_2$$

$$= 4 - 1 + 6 + 7 - 1 - 5 = 10 \quad (p, s)$$

INTEGER QUESTIONS

$$53. f(x) = \begin{cases} e^{-x}, & \text{if } x \geq 0 \\ e^x, & \text{if } x < 0 \end{cases}$$

$$f'(0^-) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{e^{-h} - 1}{-h} = 1$$

LIMITS, CONTINUITY & DIFFERENTIABILITY

$$\text{ence } \lim_{x \rightarrow 0} \frac{\left[1 - \cos \left(1 - \cos \frac{x}{2} \right) \right] \left(1 - \cos \frac{x}{2} \right)}{\left(1 - \cos \frac{x}{2} \right)^2 \cdot 2^m \cdot x^n} = 1$$

$$\text{Using } \lim_{x \rightarrow 0} \frac{1 - \cos \theta}{\theta^2} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0} \frac{\left(1 - \cos \frac{x}{2} \right)}{2^m \cdot x^n} = 2, \quad \lim_{x \rightarrow 0} \frac{4 \sin^4 \frac{x}{4}}{2^m \cdot x^n} = 2$$

$$\lim_{x \rightarrow 0} \frac{4x^4}{4^4 2^m \cdot x^n} = 2 \quad \therefore \text{for limit to exist } n = 4$$

$$\frac{2}{2^{8+m}} = 1, \quad 2^{8+m} = 2$$

$$\therefore m = -7 \Rightarrow n = 4 \text{ and } m = -7$$

$$\text{Hence } n^2 + m = 4^2 + (-7) = 16 - 7 = 9.$$

54. i) Continuity should be checked at the end-points of intervals of each definition, i.e., $x = 1, 3, 3.5, \dots$

ii) For $\{x^2\}$, continuity should be checked when $x^2 = 10, 11, 12$ or $x = \sqrt{10}, \sqrt{11}, \sqrt{12}$, $\{x^2\}$ is discontinuous for those values of x , where x^2 is an integer.

[Note: Here x^2 is monotonic for the given domain.]

iii) For $\text{sgn}(x-2)$, continuity should be checked when $x-2 = 0$ or $x = 2$

iv) For $[\log_e x]$, continuity should be checked when $\log_e x = 1$ or $x = e (\in [1, 3])$.

Hence, the overall continuity must be checked at $x = 1, 2, e, 3, \sqrt{10}, \sqrt{11}, \sqrt{12}, 3.5$.

Further, $f(1) = 0$ and

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \text{sgn}(x-2) \times [\log_e x] = 0.$$

Hence $f(x)$ is continuous at $x=1$,

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \operatorname{sgn}(x-2) \times [\log_e x] = (-1) \times 0 = 0$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \operatorname{sgn}(x-2) \times [\log_e x] = (1) \times 0 = 0$$

Hence, $f(x)$ is continuous at $x=2$,

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} \operatorname{sgn}(x-2) \times [\log_e x] = 1$$

$$\Rightarrow \lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} \{x^2\} = 0$$

Hence, $f(x)$ is discontinuous at $x=3$.

Also $\{x^2\}$ is discontinuous at

$$x = \sqrt{10}, \sqrt{11}, \sqrt{12},$$

Therefore,

$$\lim_{x \rightarrow 3.5^-} f(x) = \lim_{x \rightarrow 3.5^-} \{x^2\} = 0.25 = f(3.5).$$

Hence, $f(x)$ is discontinuous at

$$x = 3, \sqrt{10}, \sqrt{11}, \sqrt{12}.$$

55. Putting $x = 1 + h$, we get

$$\begin{aligned} f(a) &= \lim_{h \rightarrow 0} \frac{(1+h)^a - a(1+h) + a - 1}{h^2} \\ &= \lim_{h \rightarrow 0} \frac{\left(1 + ah \frac{a(a-1)}{2!} h^2 + \dots\right) - a - ah + a - 1}{h} \\ &= \frac{a(a-1)}{2} \end{aligned}$$

Therefore $f(4) = 6$.

56 & 57. Conceptual

58. Let

$$\begin{aligned} S &= \sum_{r=1}^{\infty} \frac{r+2}{2^{r+1} \cdot r \cdot (r+1)} = \sum_{r=1}^{\infty} \frac{2(r+1) - r}{2^{r+1} \cdot r \cdot (r+1)} \\ &= \sum_{r=1}^{\infty} \frac{1}{2^{r+1}} \left(\frac{2}{r} - \frac{1}{r+1} \right) = \sum_{r=1}^{\infty} \left(\frac{1}{2^r \cdot r} - \frac{1}{2^{r+1} (r+1)} \right) \end{aligned}$$

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \left[\left(\frac{1}{2^1 \cdot 1} - \frac{1}{2^2 \cdot 2} \right) + \left(\frac{1}{2^2 \cdot 2} - \frac{1}{2^3 \cdot 3} \right) + \left(\frac{1}{2^3 \cdot 3} - \frac{1}{2^4 \cdot 4} \right) \right. \\ &\quad \left. + \dots + \left(\frac{1}{2^n \cdot n} - \frac{1}{2^{n+1} \cdot (n+1)} \right) \right] \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{2} - \frac{1}{2^{n+1} \cdot (n+1)} \right) \\ \therefore S &= \frac{1}{2} \quad \text{Hence } S^{-1} = 2 \end{aligned}$$

$$59. \lim_{x \rightarrow 1} \left[\frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right]^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$$

$$\lim_{x \rightarrow 1} \left[\frac{\frac{\sin(x-1)}{(x-1)} - a}{\frac{\sin(x-1)}{(x-1)} + 1} \right] = \frac{1}{4} \Rightarrow \left(\frac{1-a}{2} \right)^2 = \frac{1}{4}$$

$$\Rightarrow a = 0, a = 2 \Rightarrow a = 2$$

60. It is obvious n is even, then

$$\begin{aligned} &\lim_{n \rightarrow \infty} \left(2^{1+3+5+\dots+n/2 \text{ terms}} \times 3^{2+4+6+\dots+n/2 \text{ terms}} \right)^{\frac{1}{(n^2+1)}} \\ &= \lim_{n \rightarrow \infty} \left(2^{\frac{n^2}{4}} \times 3^{\frac{n(n+2)}{4}} \right)^{\frac{1}{(n^2+1)}} \\ &= \lim_{n \rightarrow \infty} 2^{\frac{n^2}{4(n^2+1)}} \times 3^{\frac{n(n+2)}{4(n^2+1)}} \end{aligned}$$

$$= 2^{\lim_{n \rightarrow \infty} \frac{1}{4 \left(1 + \frac{1}{n^2} \right)}} \times 3^{\lim_{n \rightarrow \infty} \frac{\left(1 + \frac{2}{n} \right)}{4 \left(1 + \frac{1}{n^2} \right)}} = 2^{\frac{1}{4}} 3^{\frac{1}{4}} = (6)^{\frac{1}{4}}$$

$$\begin{aligned} 61. \lim_{x \rightarrow 0} \frac{\sin x^4 - x^4 \cos x^4 + x^{20}}{x^4 (e^{2x^4} - 1 - 2x^4)} \\ = \lim_{t \rightarrow 0} \frac{\sin t - t \cos t + t^5}{t (e^{2t} - 1 - 2t)} \end{aligned}$$

$$= \lim_{t \rightarrow 0} \frac{t - \frac{t^3}{3!} + \frac{t^5}{5!} \dots - t \left(1 - \frac{t^2}{2!} + \frac{t^4}{4!} \dots \right) + t^5}{t \left(1 + 2t + \frac{4t^2}{2!} + \frac{8t^3}{3!} + \frac{16t^4}{4!} + \dots - 1 - 2t \right)}$$

$$= \lim_{t \rightarrow 0} \frac{-\frac{t^3}{6} + \frac{t^3}{2} + \frac{t^5}{5!} - \frac{t^5}{4!} + \dots + t^5}{2t^3 + \frac{8t^4}{3!} + \dots}$$

$$= \frac{-\frac{1}{6} + \frac{1}{2}}{2} = \frac{-1+3}{12} = \frac{1}{6}$$

$$62. \quad \lim_{x \rightarrow 0} \sin \left(\frac{\pi(1 - \cos^m x)}{x^n} \right)$$

$$= \sin \left(\lim_{x \rightarrow 0} \frac{\pi(1 - \cos^m x)}{x^n} \right)$$

$$= \sin \left(\lim_{x \rightarrow 0} 2\pi m \frac{\sin^2 x / 2}{x^n} \right)$$

$$\Rightarrow m \in \mathbb{N} \text{ and } n = 1 \text{ or } 2$$

$$63. \quad L = e^{\lim_{x \rightarrow \infty} \frac{\ln(e^{2x} + e^x + x)}{x}}$$

using L'Hospital's rule

$$L = e^{\lim_{x \rightarrow \infty} \frac{2e^{2x} + e^x + 1}{e^{2x} + e^x + x}} = e^{\lim_{x \rightarrow \infty} \frac{2 + e^{-x} + e^{-2x}}{1 + e^{-x} + x e^{-2x}}} = e^2$$

$$64. \quad \sin^{-1} x \text{ is a monotonically increasing function. Hence, } f(x) = [\sin^{-1} x] \text{ is discontinuous, where } \sin^{-1} x \text{ is an integer.}$$

$$\Rightarrow \sin^{-1} x = -1, 0, 1 \text{ or } x = -\sin 1, 0, \sin 1$$

$$\frac{2}{1+x^2}, x \geq 0, \text{ is a monotonically decreasing function.}$$

$$\text{Hence, } f(x) = \left[\frac{2}{1+x^2} \right], x \geq 0 \text{ is discontinuous when } \frac{2}{1+x^2} \text{ is an integer.}$$

$$\Rightarrow \frac{2}{1+x^2} = 1, 2 \Rightarrow x = 1, 0$$

$$65. \quad \sin(x-2) \text{ and } \cos(x-2) \text{ are continuous}$$

for all x . Since $[x^3]$ is not continuous at integral values of x^3 , $f(x)$ is continuous in

$$[0, 2] \text{ if } \left[\frac{(x-2)^3}{a} \right] = 0, \forall x \in [0, 2].$$

$$\text{Now, } (x-2)^3 \in [0, 8] \text{ for } x \in [4, 6]$$

$$\Rightarrow a > 8 \text{ for } \left[\frac{(x-2)^3}{a} \right] = 0$$

$$66. \quad f(x) = \begin{cases} x-1, & x \text{ is rational} \\ x^2-x-2, & x \text{ is irrational} \end{cases} \text{ is}$$

continuous when $x-1 = x^2-x-2$

$$\text{or } x^2-2x-1=0 \text{ or } x = \frac{2 \pm \sqrt{8}}{2}.$$

Hence, $f(x)$ is continuous at two points

$$\Rightarrow a = 2.$$

$f(x) = \operatorname{sgn}(x^3 - 3x + 1)$ is discontinuous when $x^3 - 3x + 1 = 0$.

Now, $y = x^3 - 3x + 1$ has three distinct roots

$$\text{as } \frac{dy}{dx} = 3x^2 - 3 = 0 \Rightarrow x \pm 1$$

$$\text{Also } f(1) = 1 - 3 + 1 = -1 \text{ and}$$

$$f(-1) = -1 + 3 + 1 = 3. \text{ Hence, the graph of } y = x^3 - 3x + 1 \text{ is}$$

Hence, $f(x) = \operatorname{sgn}(x^3 - 3x + 1)$ is discontinuous at three points $\Rightarrow b = 3$.

$$f(x) = (\log x) |x^2 - 4x + 3| + 2(x-2)^{1/3} \\ = (\log x) |x-1| |x-3| + 2(x-2)^{1/3}$$

Which is non-differentiable at $x = 3$ and

$$x = 2 \text{ as at } x = 3, f(x) \text{ has a sharp turn}$$

and at $x = 2$, $f(x)$ have a vertical tangent.

At $x = 1$, $f(x)$ is differentiable. So $c = 2$

and $d = 1$

$$67. \quad f(x) = \lim_{n \rightarrow \infty} \frac{(x^2)^n - 1}{(x^2)^n + 1}$$

$$= \lim_{n \rightarrow \infty} \frac{1 - \frac{1}{(x^2)^n}}{1 + \frac{1}{(x^2)^n}}$$

$$= \begin{cases} 1, & x < -1 \\ -1, & 0 \leq x^2 < 1 \\ 0, & x^2 = 1 \\ 1, & x^2 > 1 \end{cases} = \begin{cases} 1, & x < -1 \\ 0, & x = -1 \\ -1, & -1 < x < 1 \\ 0, & x = 1 \\ 1, & x > 1 \end{cases}$$

Thus $f(x)$ is discontinuous at $x = \pm 1$. 69.

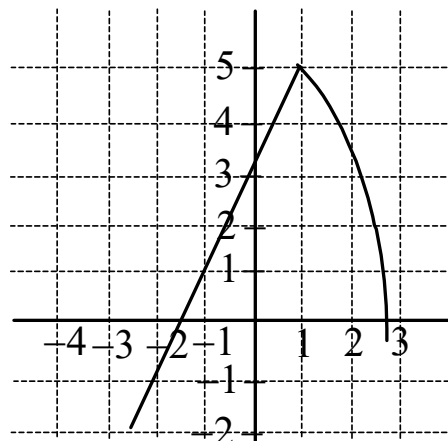
$$f(x) = \lim_{n \rightarrow \infty} \frac{(x^2)^n - 1}{(x^2)^n + 1}$$

$$= \lim_{n \rightarrow \infty} \frac{1 - \frac{1}{(x^2)^n}}{1 + \frac{1}{(x^2)^n}}$$

$$= \begin{cases} 1, & x < -1 \\ -1, & 0 \leq x^2 < 1 \\ 0, & x^2 = 1 \\ 1, & x^2 > 1 \end{cases} = \begin{cases} 1, & x < -1 \\ 0, & x = -1 \\ -1, & -1 < x < 1 \\ 0, & x = 1 \\ 1, & x > 1 \end{cases}$$

Thus $f(x)$ is discontinuous at $x = \pm 1$.

Graph of $y = f(x)$



68.

Graph of $y = f(|x|)$

